

Sparse and Deep Gaussian Processes Closure for 2-D Fluids and Ocean Flows

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Spatial Closure

Motivation and Challenges

Motivation:

Fully resolved physics:

- High accuracy at High Cost
- Need for reduction

Under-resolved physics:

- Lower accuracy at Lower Cost
- Truncation / discretization error

Challenges:

- Complex nonlinear, multiscale fluid dynamics with unresolved sub-grid physics
- Truncated numerical models introduce truncation and discretization biases
- Regime variability (Reynolds-number shifts, transient forcing) and evolving flow structures

Needs:

- Fast, online data-driven correction of low-resolution dynamics, embedded directly into the solver
- Uncertainty-aware predictions of resolved fields and wake metrics
- Transferable across resolutions, domains, and flow conditions

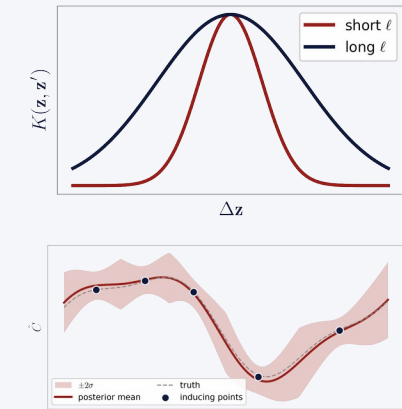
Deep Gaussian Process

SVGP on deep features

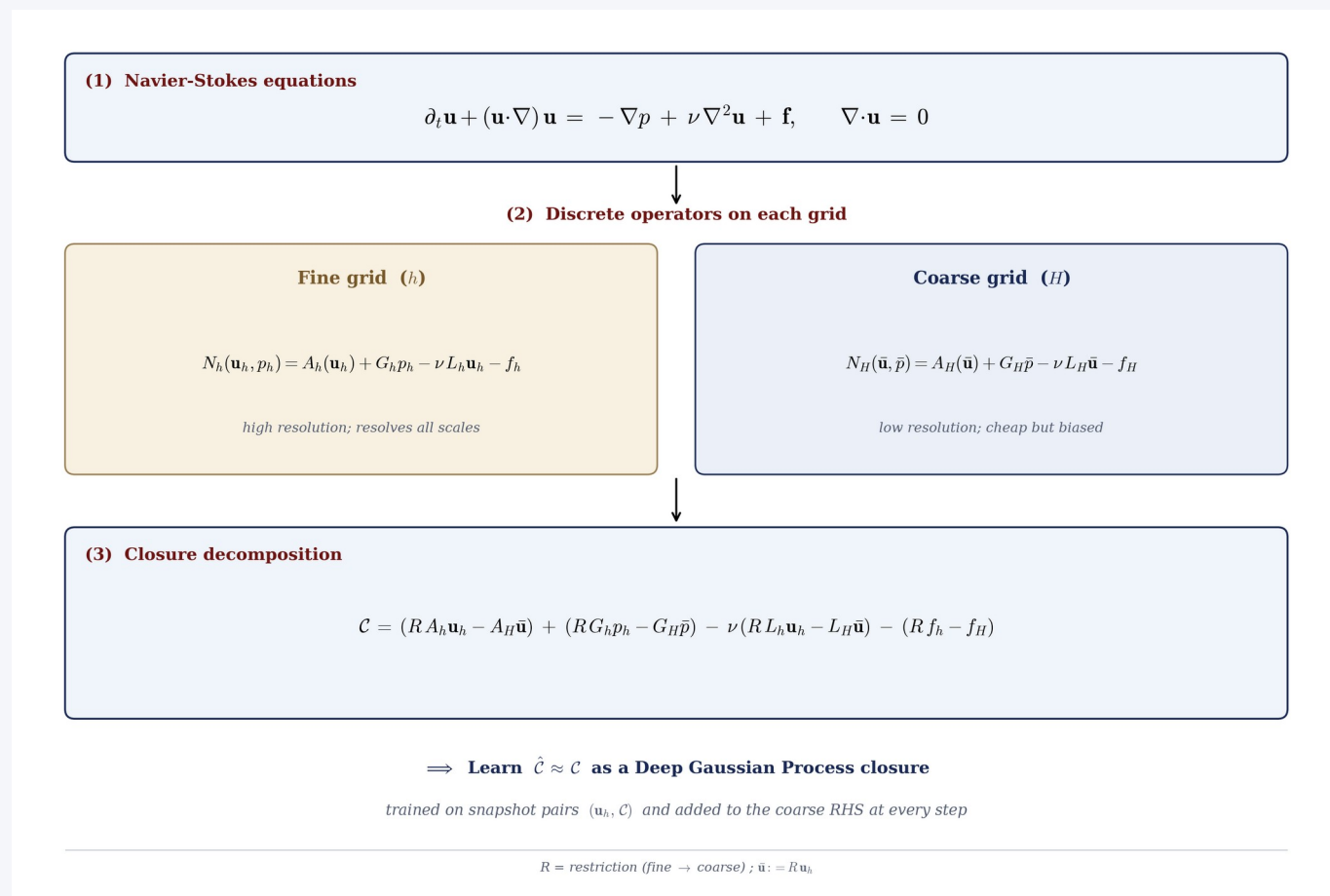
$$\mathbf{z}_{i,j} = \text{CNN}(\Phi)_{i,j}$$

$$\hat{C}_{i,j} \sim \mathcal{GP}(\mu(\mathbf{z}_{i,j}), K(\mathbf{z}_{i,j}, \mathbf{z}'_{i,j}))$$

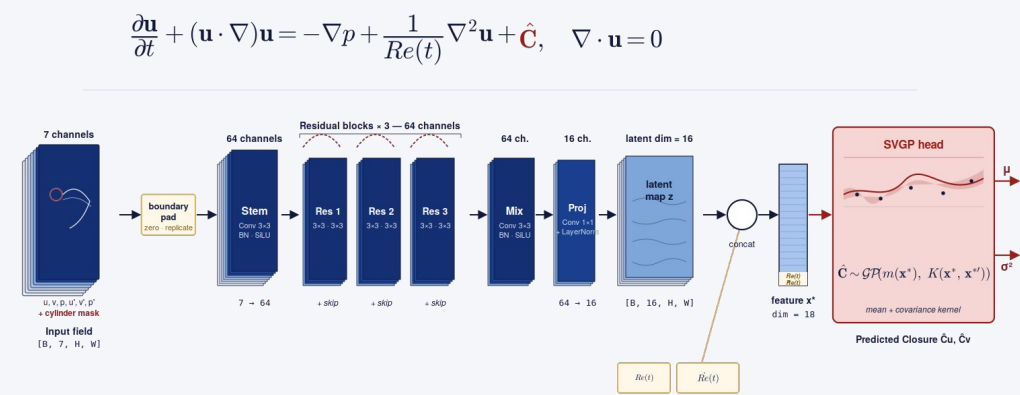
- Structured priors via the kernel (smoothness, length-scales)
- Non-parametric — no rigid analytic form
- Built-in uncertainty: posterior σ^2 per pixel
- Inducing points + ELBO scale to many DOFs



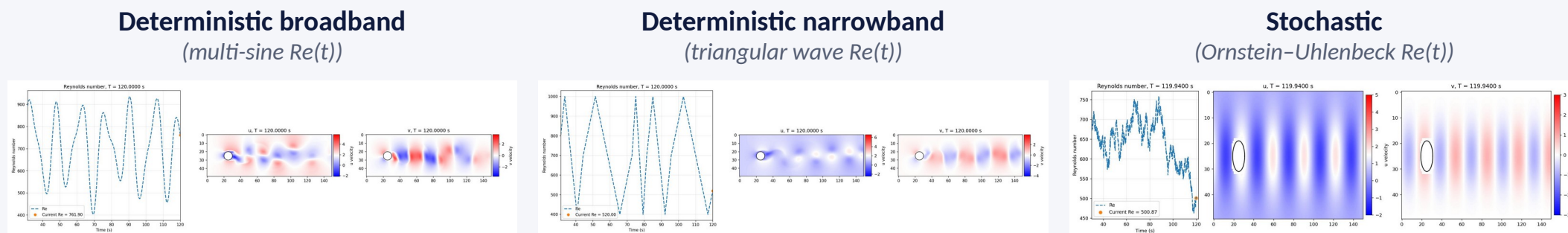
Methodology



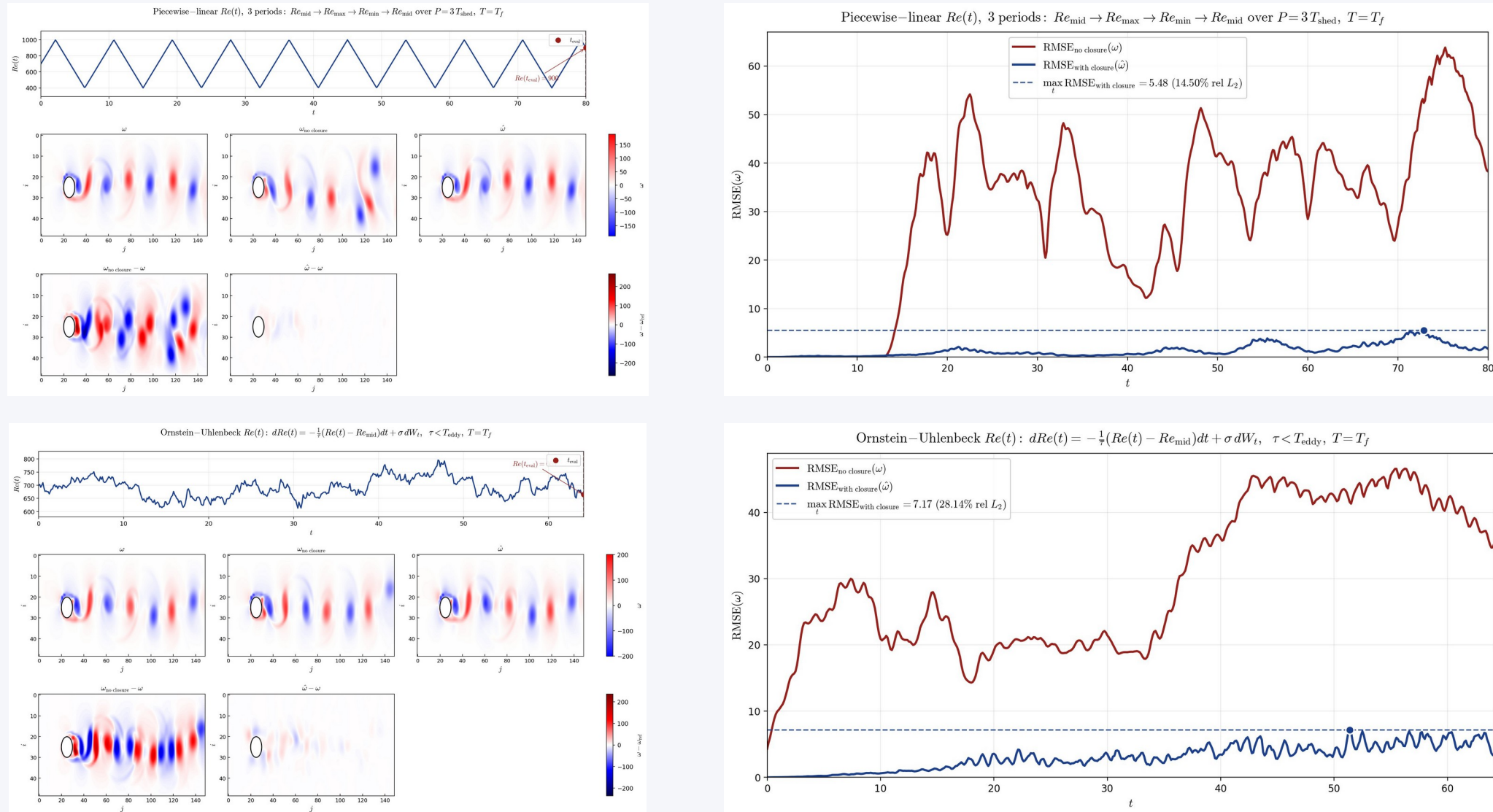
Architecture



Training: Re(t) Modulation Families



Inference Results: Out-Of-Distribution Rollouts



Conclusion · Acknowledgments

Conclusions

- Sparse variational GP closures, trained on snapshot pairs, inject mean (or stochastic) corrections directly into the FV solver and yield stable OOD rollouts on Re(t) modulations the model never saw at training.
- An analytical δR for AB2CN2 IMEX is derived: three pointwise-spectral terms + one NN surrogate ($\hat{\mathcal{I}}$). Empirical residuals match the predicted closure to high accuracy on flow past a cylinder.
- Together: the spatial closure mitigates grid-resolution bias; the temporal closure mitigates step-size bias. Combined, they target an end-to-end fast, accurate, uncertainty-aware coarse solver.

Temporal Closure

Motivation

Why a temporal closure?

- Spatial closure trains on snapshot pairs — no constraint on temporal coherence
- Coarse ΔT introduces its own truncation error, independent of grid resolution
- Errors at step k feed back into step $k+1$: rollouts can drift over long horizons
- Goal: a closure consistent in time that matches a fine-step reference at coarse cost

Approach:

- Match the local truncation error of K fine AB2CN2 steps ($dt = \Delta T/K$) with one coarse ΔT step
- Closed-form additive RHS correction δR : three terms are pointwise spectral, one ($\hat{\mathcal{I}}$) is a neural surrogate
- AB2CN2 at ΔT behaves like AB2CN2 at $\Delta T/K$, at K -fold lower runtime cost

Flow past cape
temporal under-resolution
[to be added]
 ω ($\Delta t = 10^{-5}$) | ω ($\Delta t = 10^{-3}$) | error

Methodology

(1) Truncation-error closure — match the exact flow (fine RK4)

$$\tau = \bar{\omega}(t_n + \Delta T) - \bar{\omega}_{\text{AB2CN2}}^{n+1} = -\frac{\Delta T^3}{12} R_3 - \frac{\Delta T^4}{24} R_4 + \mathcal{O}(\Delta T^5)$$

$$R_3 = L^3 \bar{\omega} + L^2 N + L \dot{N} - 5 \ddot{N}$$

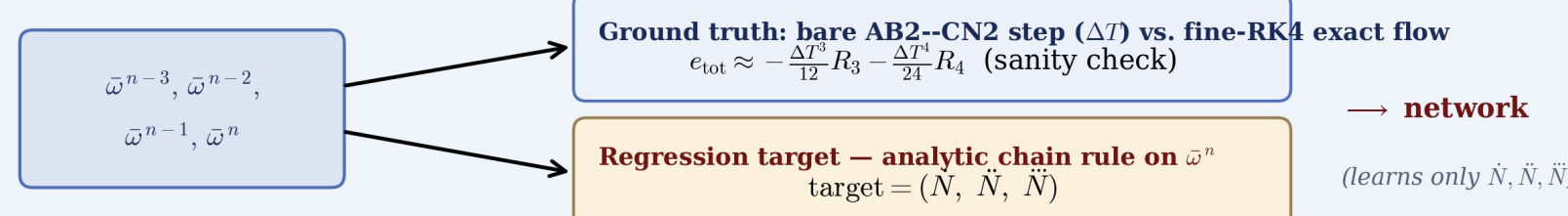
$$R_4 = 2L^4 \bar{\omega} + 2L^3 N + 2L^2 \dot{N} - 4L \ddot{N} + \ddot{\ddot{N}}$$

$$\bar{\omega}_{\text{closed}}^{n+1} = \bar{\omega}_{\text{AB2CN2}}^{n+1} - \Delta T^3 \frac{1}{12} (L^3 \bar{\omega} + L^2 N) - \Delta T^4 \frac{1}{24} (2L^4 \bar{\omega} + 2L^3 N) - f_{\text{NN}}$$

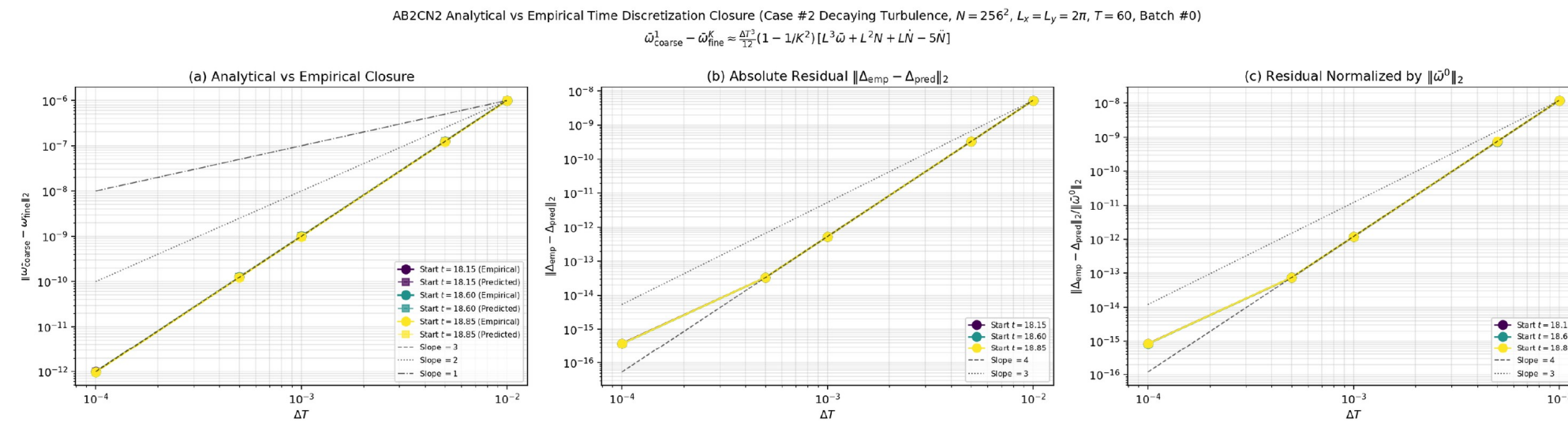
$$f_{\text{NN}} = \Delta T^3 \frac{1}{12} (L \dot{N} - 5 \ddot{N}) + \Delta T^4 \frac{1}{24} (2L^2 \dot{N} - 4L \ddot{N} + \ddot{\ddot{N}})$$

(2) Training-data construction

Per developed-flow seed — 4-level stencil ($\bar{\omega}^{n-3}, \dots, \bar{\omega}^n$) via ultra-fine RK4



Analytical / learned split : $R_p = R_p^{\text{anal}} + R_p^{\text{NN}}$
 $R^{\text{anal}}: L^k$ on $\bar{\omega}, N$ (Fourier-diagonal, free) $R^{\text{NN}}: (\dot{N}, \ddot{N}, \ddot{\ddot{N}})$ learned · per-channel rel- L^2 loss



Architecture

